

WEEK 4 PROGRESS REPORT

Week 4 Report

Prompt Engineering, Artificial Intelligence & Gemini API Integration

Weekly Objective

The objective of this week was to understand the fundamentals of Artificial Intelligence, learn Prompt Engineering techniques, explore the working of Large Language Models (LLMs), and understand how AI services can be integrated into web applications using the Google Gemini API. The week also focused on designing and implementing an AI-powered feature using WordPress, Make.com, and Gemini.

Day 1 – Understanding Artificial Intelligence

Activities Performed

- Studied the fundamentals of Artificial Intelligence (AI).
- Learned the concept of Large Language Models (LLMs) and how they generate responses based on language patterns.
- Explored the working principles of AI models such as Google Gemini and ChatGPT.
- Compared different prompts to observe how prompt wording affects AI-generated responses.
- Identified practical use cases of AI within software applications.

Learning Outcomes

- Understood that AI predicts responses based on learned language patterns rather than memorizing information.
- Learned the workflow of an LLM:
User Prompt → Context Analysis → Response Generation → Output
- Gained knowledge of practical AI applications in modern software systems.

Day 2 – Prompt Engineering

Activities Performed

- Learned the concept and importance of Prompt Engineering.
- Practiced writing structured prompts using four key components:
 - Role
 - Context
 - Task
 - Output Format
- Compared weak and well-structured prompts to evaluate response quality.
- Generated website content such as homepage text, About Us, Services, and FAQs using AI.

Learning Outcomes

- Understood that the quality of AI responses depends largely on prompt quality.
- Learned techniques for creating clear and effective prompts.
- Improved skills in generating professional website content with AI assistance.

Day 3 – Gemini API Planning

Activities Performed

- Studied the purpose and working of API Keys.
- Learned secure API key management practices.
- Explored Google AI Studio and understood how Gemini API provides AI capabilities to applications.

- Planned the AI feature for the project by defining:
 - User input
 - Expected AI response
 - Data flow
 - User interaction

Learning Outcomes

- Understood the importance of API authentication and security.
- Learned the communication flow between a web application and the Gemini API.
- Developed an implementation plan for AI integration.

Day 4 – Gemini API Workflow

Activities Performed

- Studied the architecture of AI-powered web applications.
- Learned how websites communicate with AI services through APIs.
- Explored the concept of Temperature and its effect on AI-generated responses.
- Designed the workflow for the AI feature before implementation.
- Tested multiple prompt variations to compare AI outputs.

Learning Outcomes

- Understood the complete request-response cycle between a website and an AI service.
- Learned how prompt structure and Temperature influence AI responses.
- Improved understanding of AI feature planning and workflow design.

Day 5 – AI Feature Integration

Activities Performed

- Developed an AI-powered feature for the WordPress website.
- Created a user interface using HTML and JavaScript.
- Connected the website to Make.com using a Webhook.
- Configured Google Gemini within Make.com to process user input.
- Implemented the Webhook Response module to return AI-generated results to the website.
- Performed end-to-end testing of the complete AI workflow.

Learning Outcomes

- Learned how AI can be securely integrated into web applications.
- Understood why API keys should remain hidden on the server side.
- Successfully implemented the following workflow:

WordPress Website → JavaScript → Make.com Webhook → Gemini API → Webhook Response → User Interface

Overall Weekly Learning

During Week 4, I gained practical knowledge of Artificial Intelligence, Large Language Models, Prompt Engineering, API security, and Gemini API integration. I learned how AI-powered web applications communicate with external AI services while keeping sensitive credentials secure. Furthermore, I successfully understood the complete workflow of integrating AI into a web application using WordPress, Make.com, and Google Gemini. This experience enhanced my understanding of modern AI-enabled software development and demonstrated how intelligent features can improve user interaction within web-based systems.